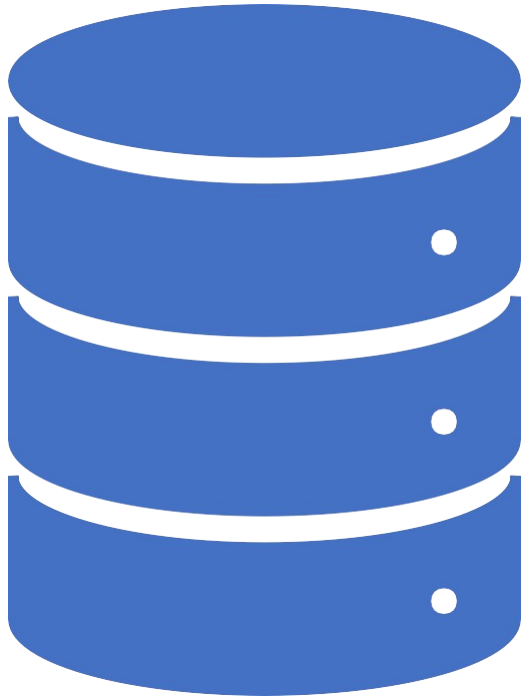
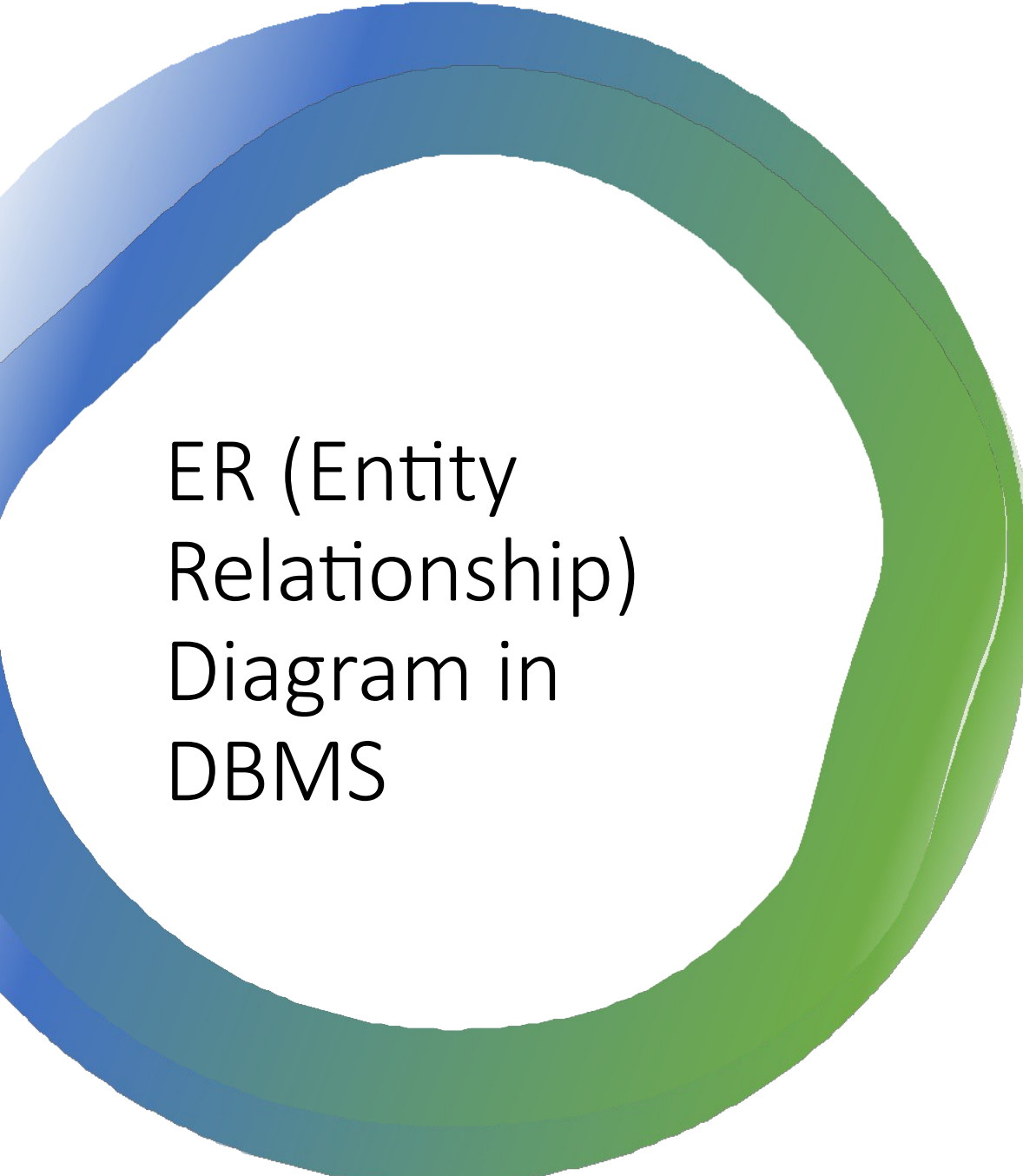




Database Systems

[ER Diagram Model in DBMS]

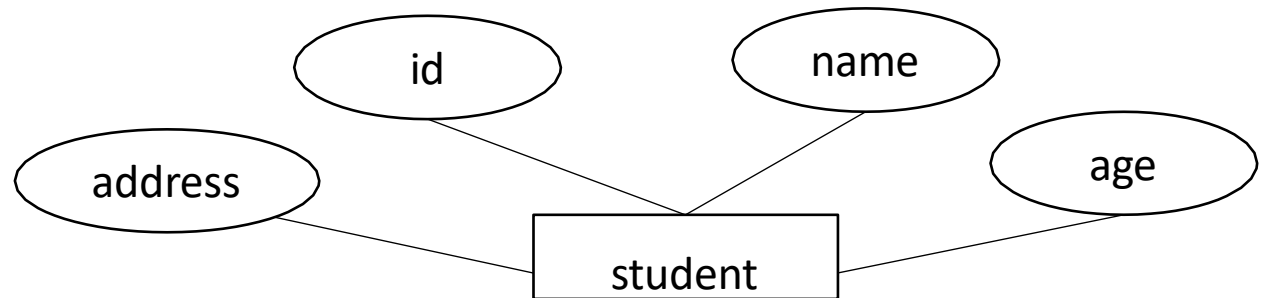


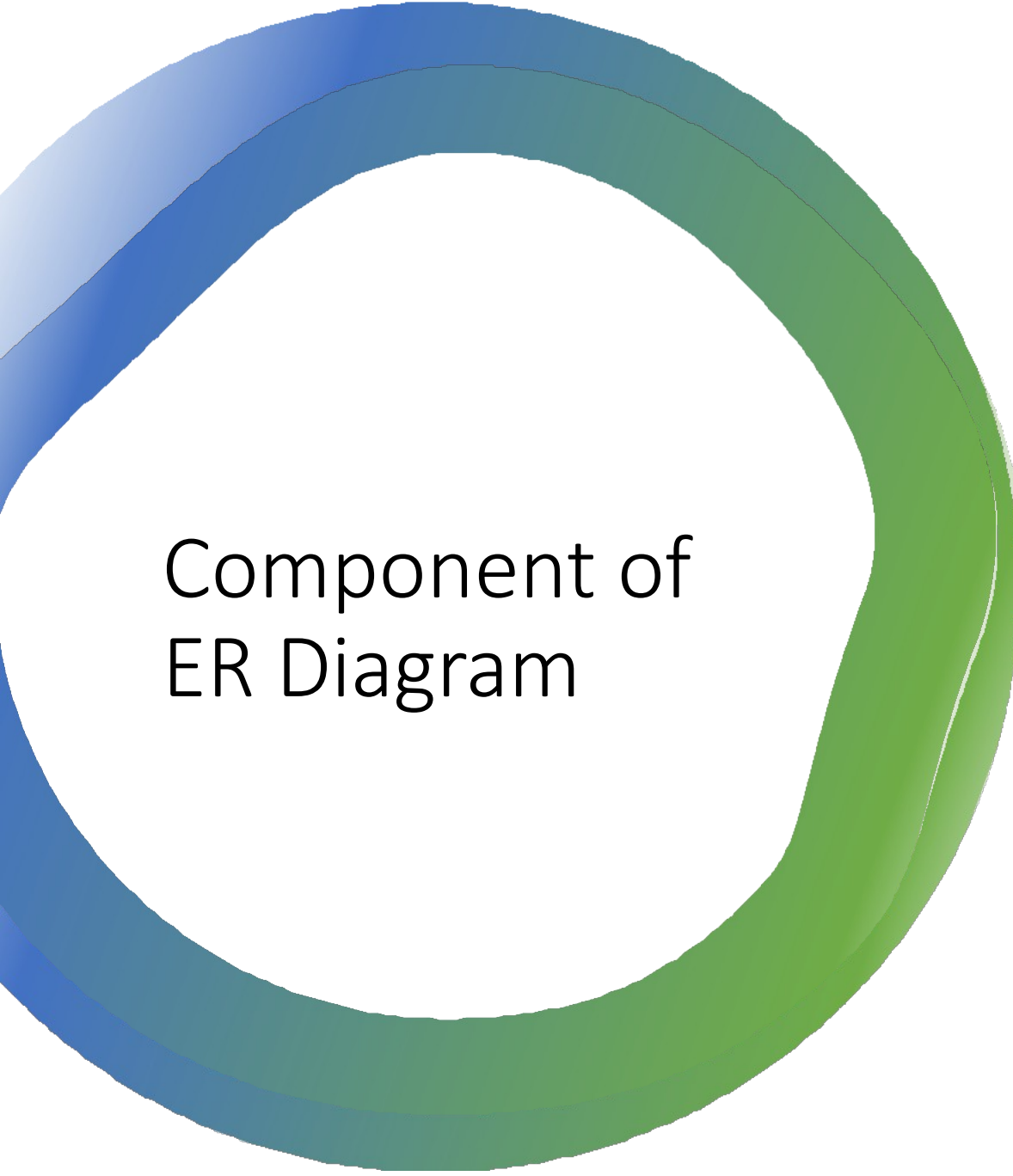
ER (Entity Relationship) Diagram in DBMS

- ER model stands for an **Entity-Relationship** model.
- It is a high-level data model. This model is used to define the data elements and relationship for a specified system.
- It develops a conceptual design for the database.
- It also develops a very simple and easy to design view of data.
- In ER modeling, the database structure is portrayed as a diagram called an **entity-relationship diagram**.

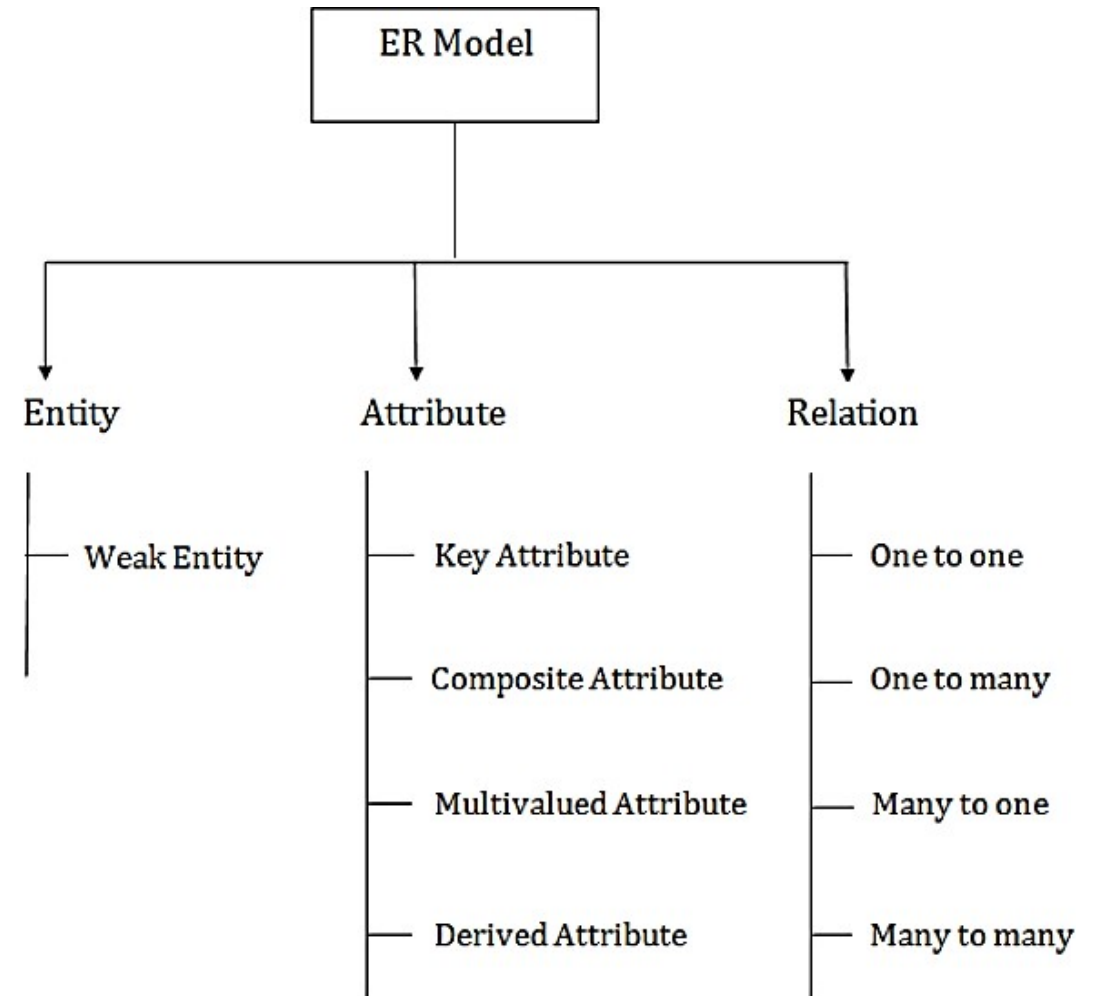
ERD in DBMS

- Suppose we design a school database.
- In this database, the student will be an entity with attributes like address, name, id, age, etc.
- The address can be another entity with attributes like city, street name, pin code, etc and there will be a relationship between them.



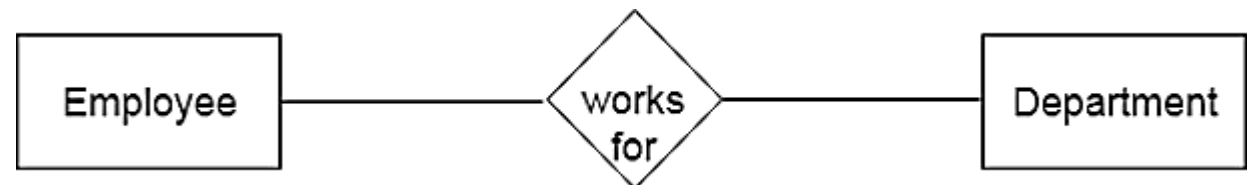


Component of ER Diagram



Entity/ Object

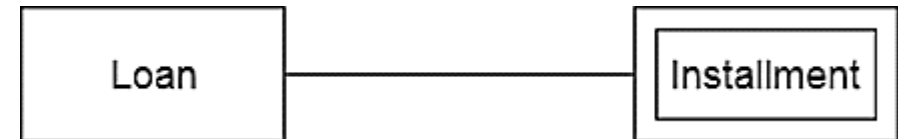
- An **entity** may be any object, class, person or place. OR anything that is participating in the system.
- In the ER diagram, an entity can be represented as ***rectangles***.
- Consider an organization as an example- manager, product, employee, department etc. can be taken as an entity.





Weak Entity

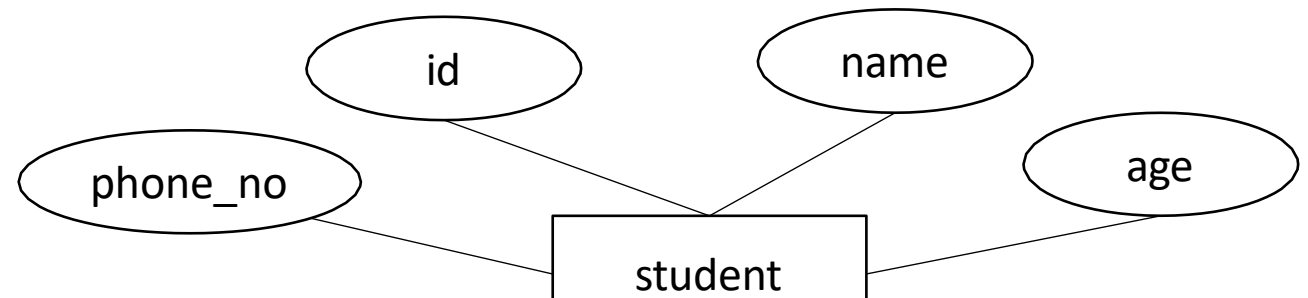
- An entity that depends on another entity called a **weak entity**.
- The weak entity doesn't contain any key attribute of its own.
- The weak entity is represented by a ***double rectangle***.





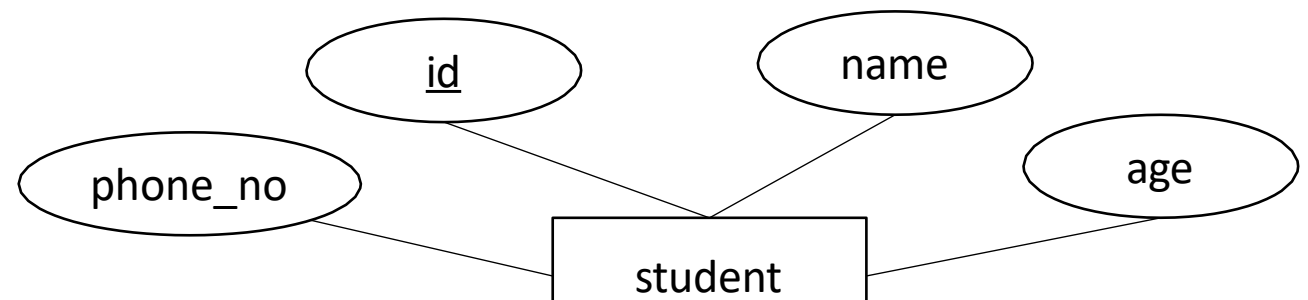
Attribute/ Properties

- The attribute is used to describe the property (characteristics) of an entity.
Eclipse is used to represent an attribute.
- For example, id, age, contact number, name, etc. can be attributes of a student.



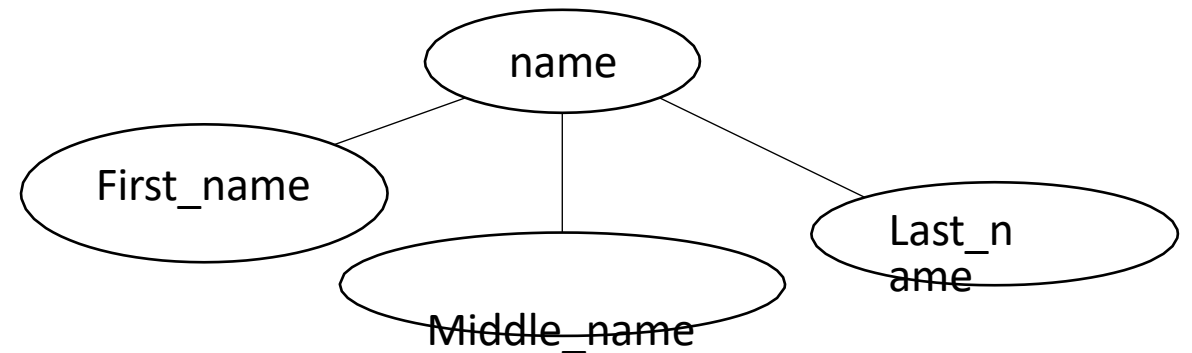
Key Attribute

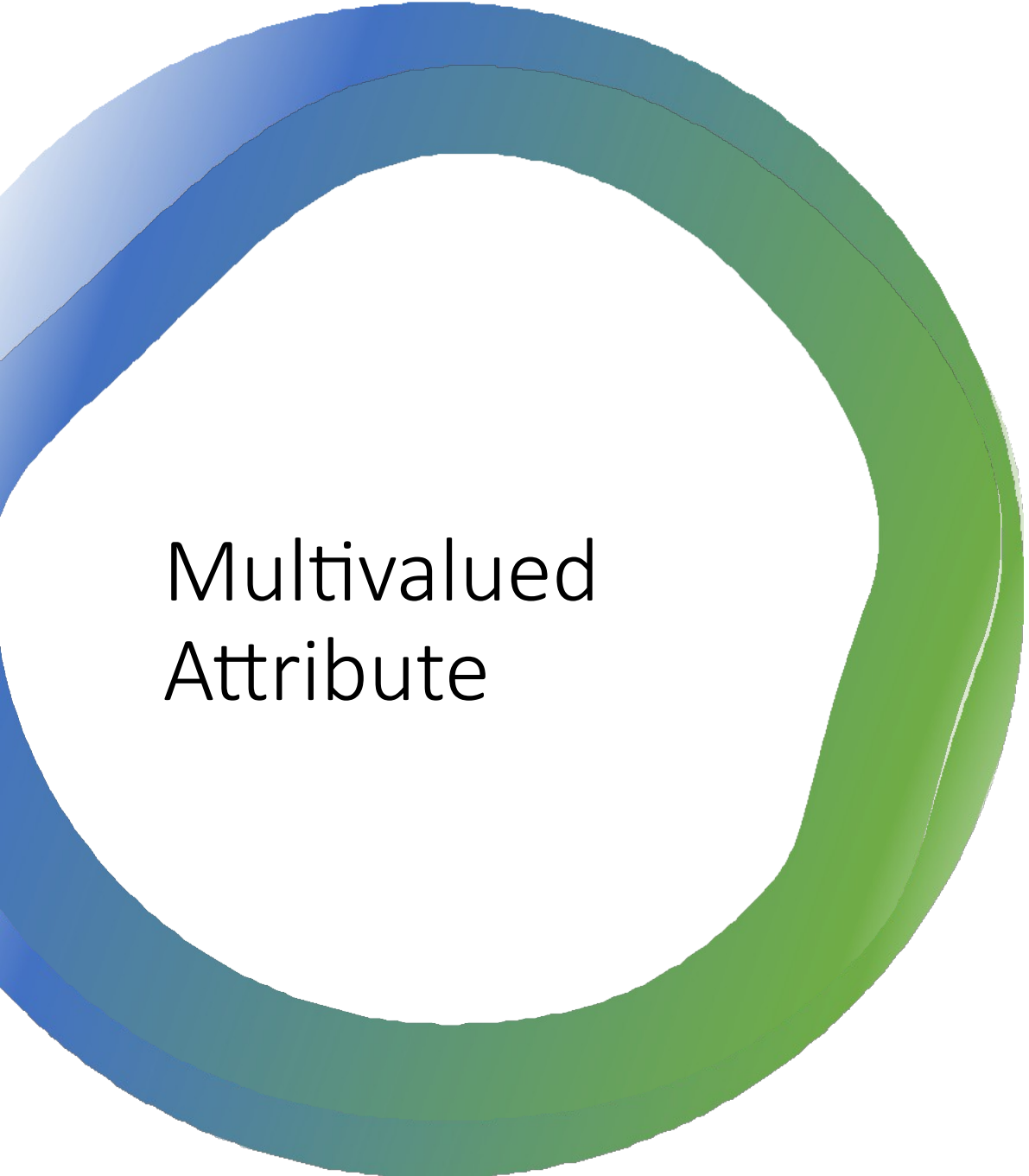
- The key attribute is used to represent the main characteristics of an entity.
- It represents a ***primary key***.
- The key attribute is represented by an ellipse with the ***text underlined***.



Composite Attribute

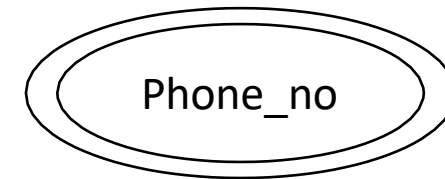
- An attribute that composed of many other attributes is known as a **composite attribute**.
- The composite attribute is represented by an ellipse, and those ellipses are connected with an ellipse.





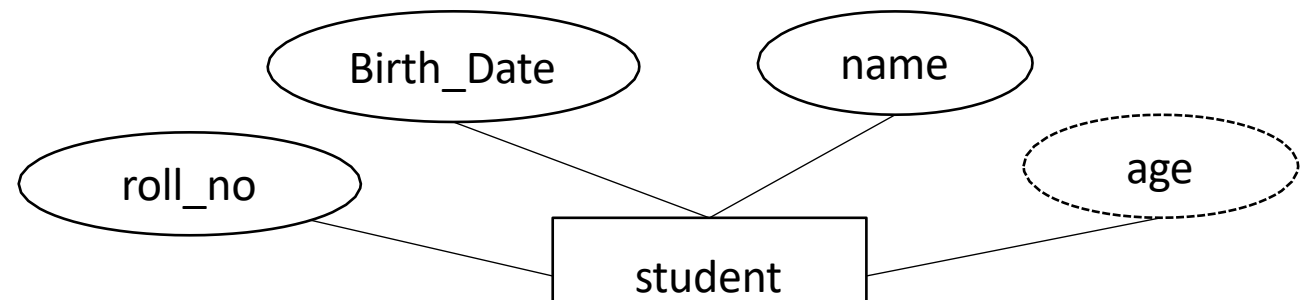
Multivalued Attribute

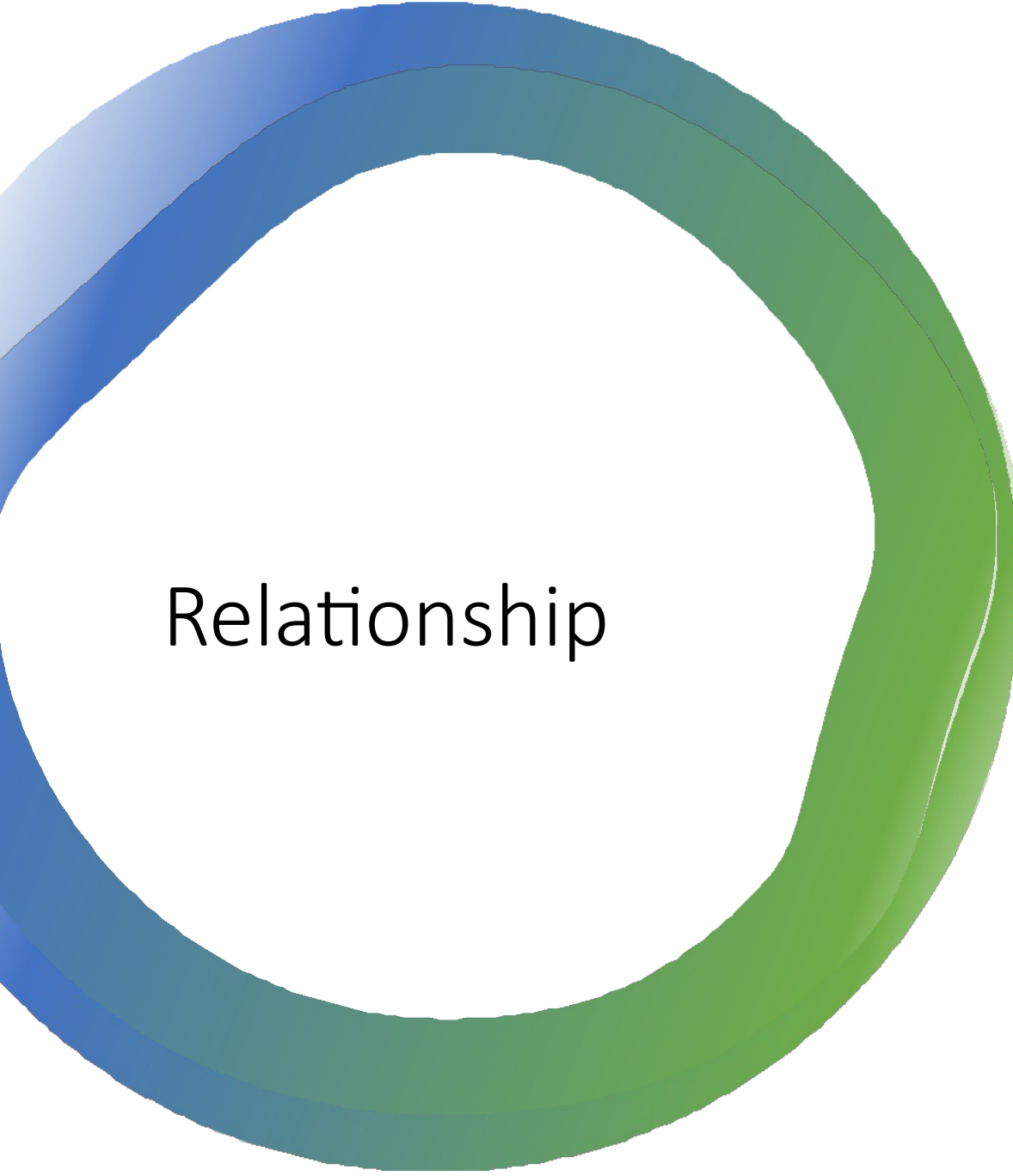
- An attribute can have more than one value. These attributes are known as a **multivalued attribute**.
- The **double oval** is used to represent multivalued attribute.



Derived Attribute

- An attribute that can be derived from other attribute is known as a ***derived attribute***.
- It can be represented by a dashed ellipse.
- For example, A person's **age** changes over time and can be derived from another attribute like *Date of birth*.





Relationship

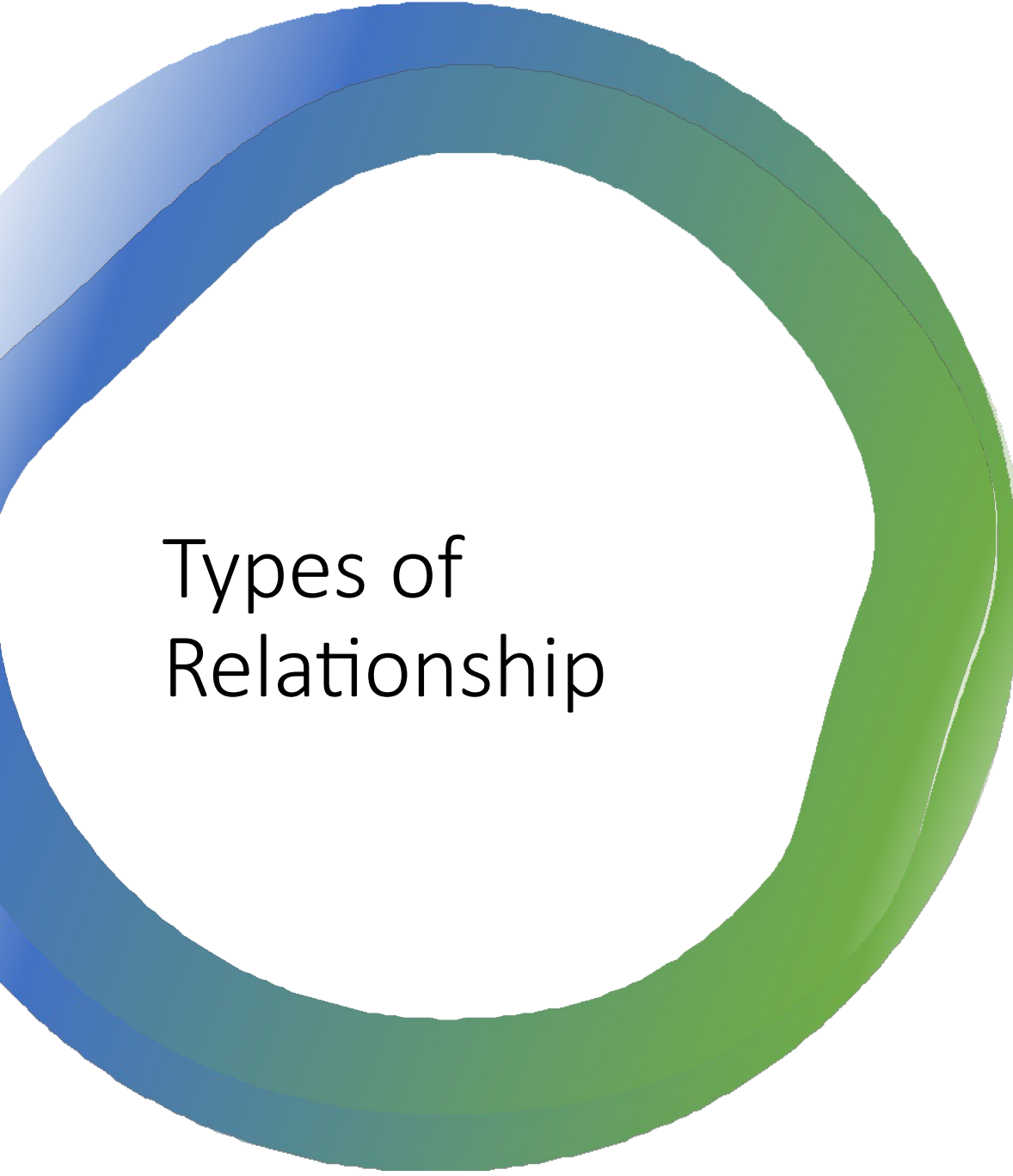
- A **relationship** is used to describe the relation between entities. OR a logical connection between different entities.
- The relationship indicates how the entities are connected or related to each other.
- The entities that participate in a relationship are **participants**.
- ***Diamond*** or ***rhombus*** is used to represent the relationship.



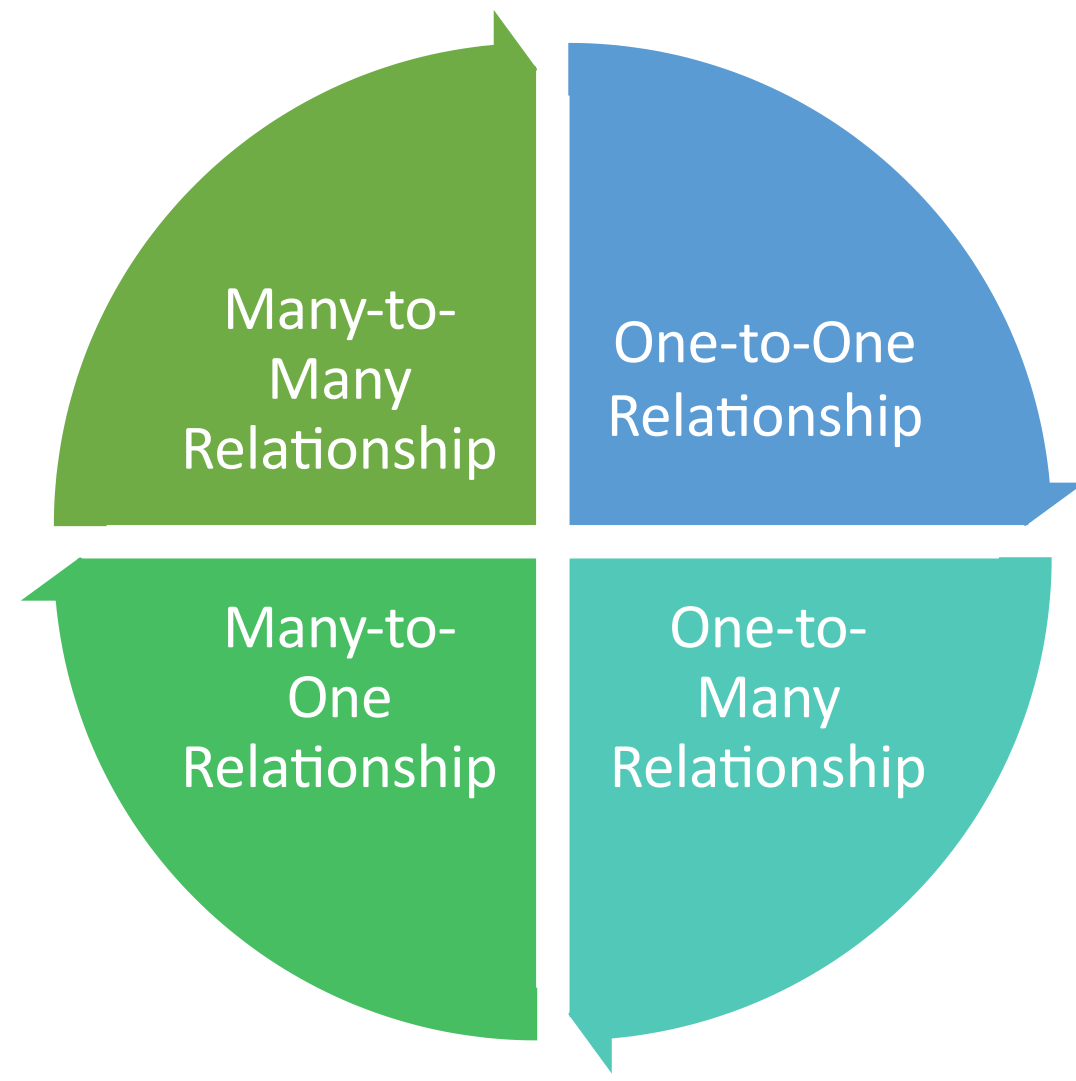


Entity Class and Entity Instance

- A group of entities of the same type is called **entity class**.
- All entities in an entity type share common characteristics.
 - For example, STUDENT entity class is a collection all students.
- A member of an entity class is also known as an **entity instance**.
 - For example, a student Abdullah of STUDENT entity type is an entity instance.



Types of Relationship



Many-to-Many Relationship

One-to-One Relationship

Many-to-One Relationship

One-to-Many Relationship

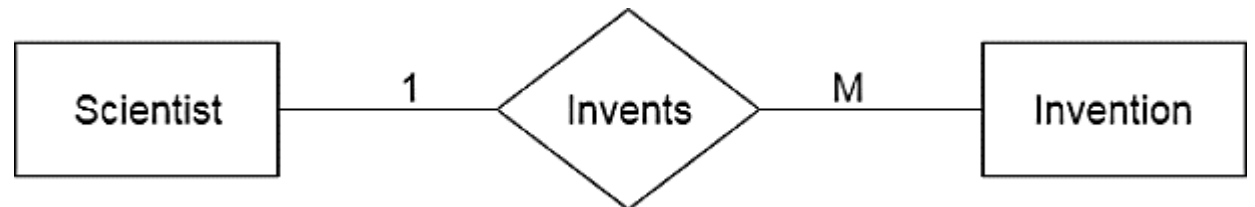
Types of Relationship (One-to-One Relationship)

- When only one instance of an entity is associated with the relationship, then it is known as ***one to one relationship***.
- For example, A female can marry to one male, and a male can marry to one female.



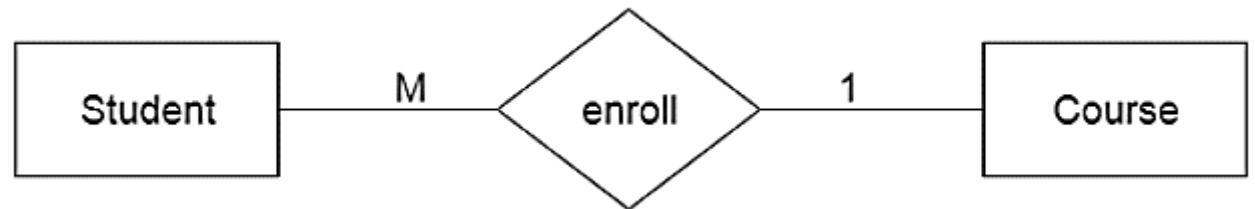
Types of Relationship (One-to-Many Relationship)

- When only one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then this is known as a ***one-to-many relationship***.
- For example, Scientist can invent many inventions, but the invention is done by the only specific scientist.



Types of Relationship (Many-to-One Relationship)

- When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as a ***many-to-one relationship***.
- For example, Student enrolls for only one course, but a course can have many students.



Types of Relationship (Many-to-Many Relationship)

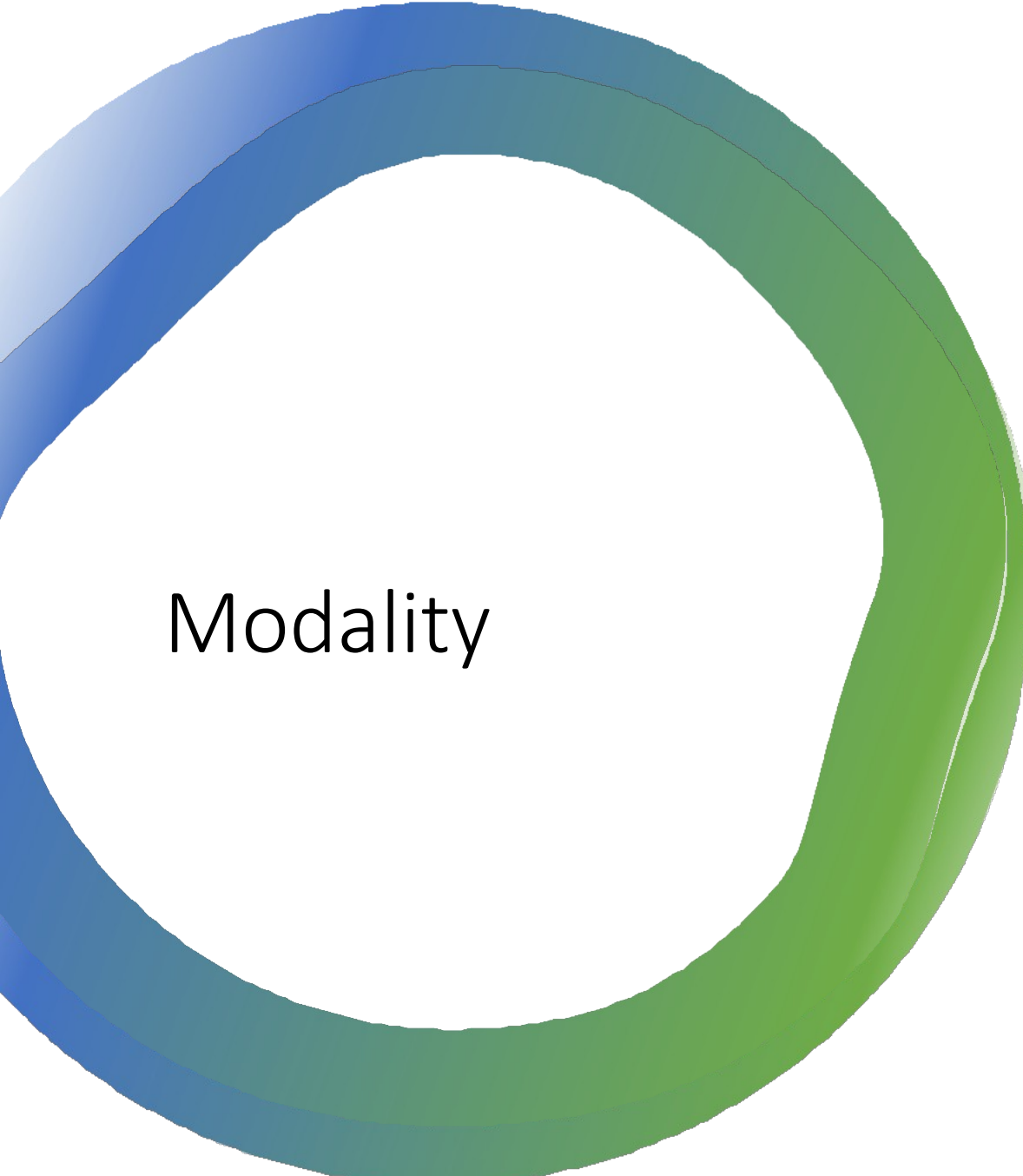
- When more than one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then it is known as a ***many-to-many relationship***.
- For example, Employee can assign by many projects and project can have many employees.





Cardinality

- The maximum number of instances of one entity associated with each instance of the related entity is known as **cardinality**.
- The cardinality can be **one** or **many**.
- The cardinality of **one** indicates **single** instance of an entity. The symbol of a **vertical line** | is used for **one** cardinality.
- The cardinality of **many** indicates **multiple** instance of an entity. The symbol of a **crow's foot** $\Leftarrow$ is used for **many** cardinality.



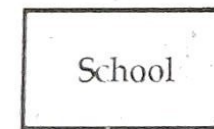
Modality

- The minimum number of instances of one entity associated with each instance of the related entity is known as **modality**.
- It describes the relationship as **mandatory** or **optional**.
 - The relationship is called **optional** when the minimum number is zero.
 - The **small circle o** is used to indicate an optional relationship.
 - The relationship is called **mandatory** when the minimum number is one.
 - The **vertical line |** is used to indicate a mandatory relationship.

Cardinality and Modality

Cardinality:

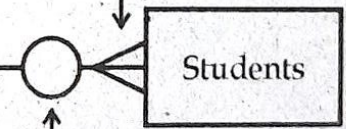
Indicates that only one school is involved in the relationship



enrolls

Cardinality:

Indicates that there may be many students in school



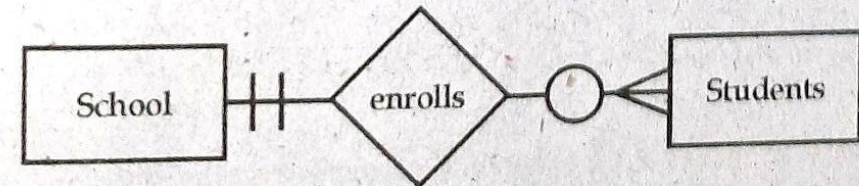
Modality: Mandatory

Indicates that there must be one school to have this relationship

Modality: Optional

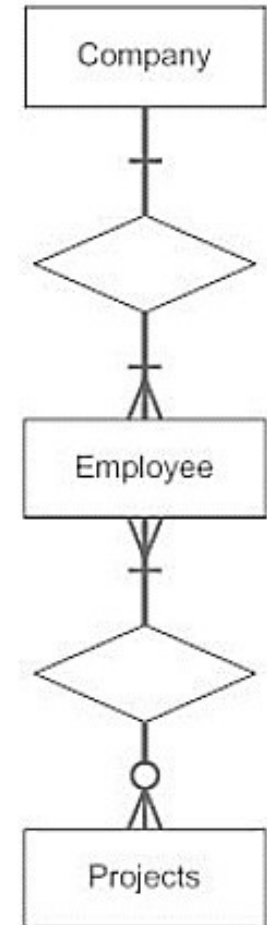
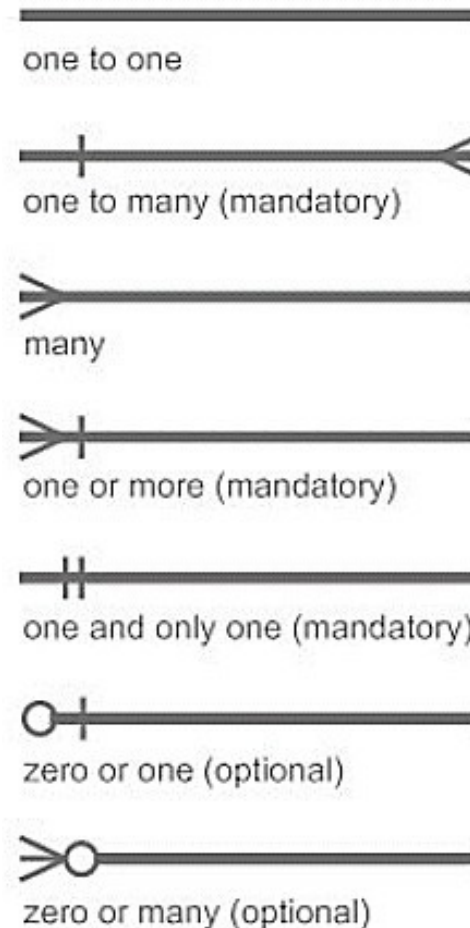
Indicates that there may be no students in school

The above figure can also be drawn as follows:



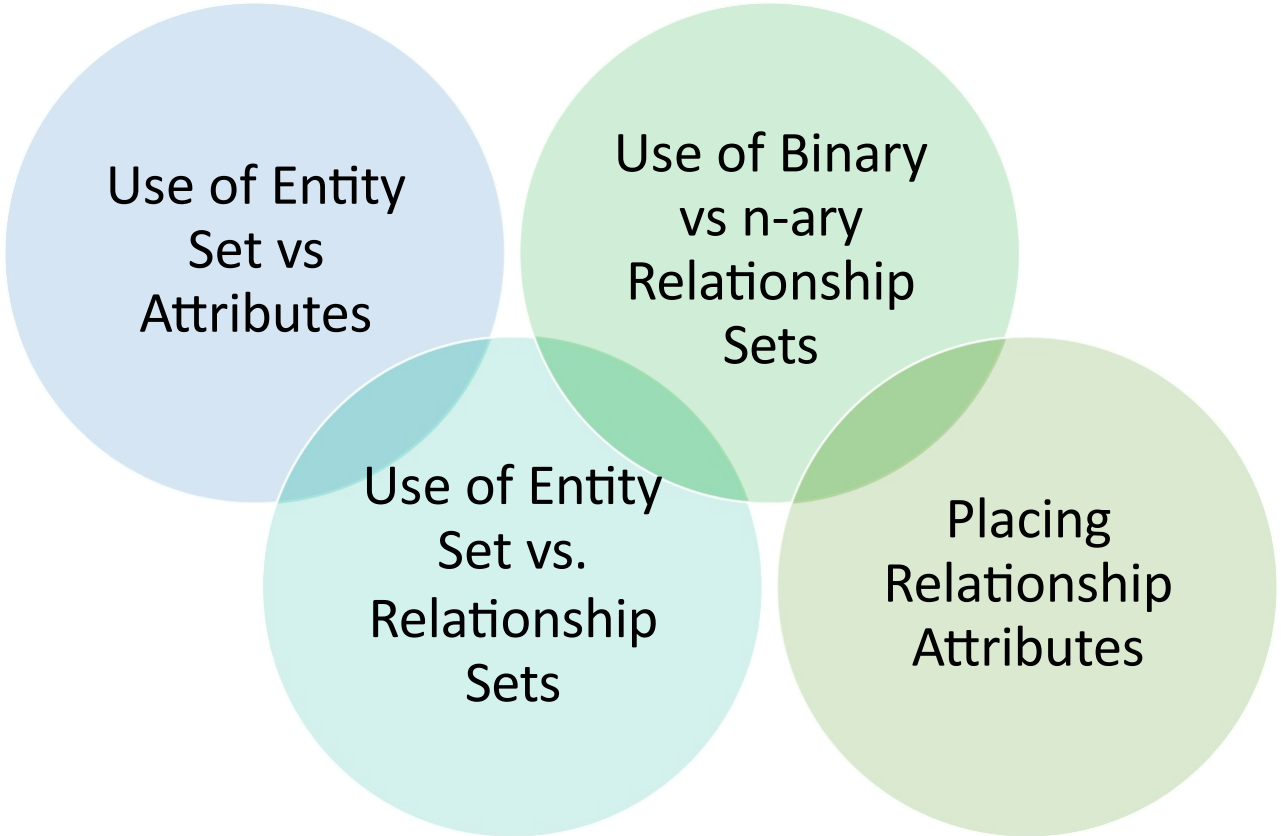
Notation of ER Diagram

- Database can be represented using the notations. In ER diagram, many notations are used to express the cardinality.
- These notations are as follows:





ER Design Issues

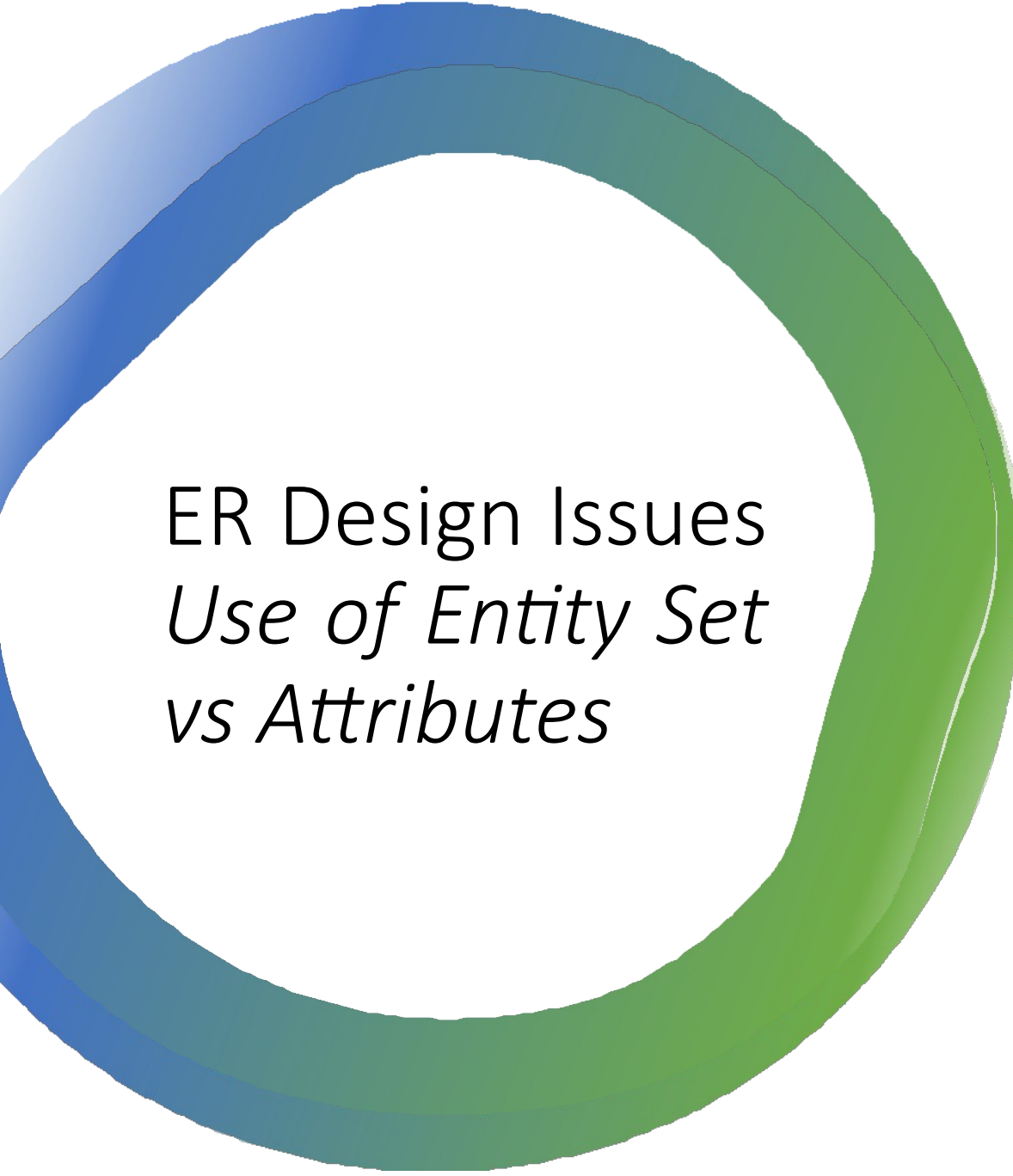


Use of Entity
Set vs
Attributes

Use of Binary
vs n-ary
Relationship
Sets

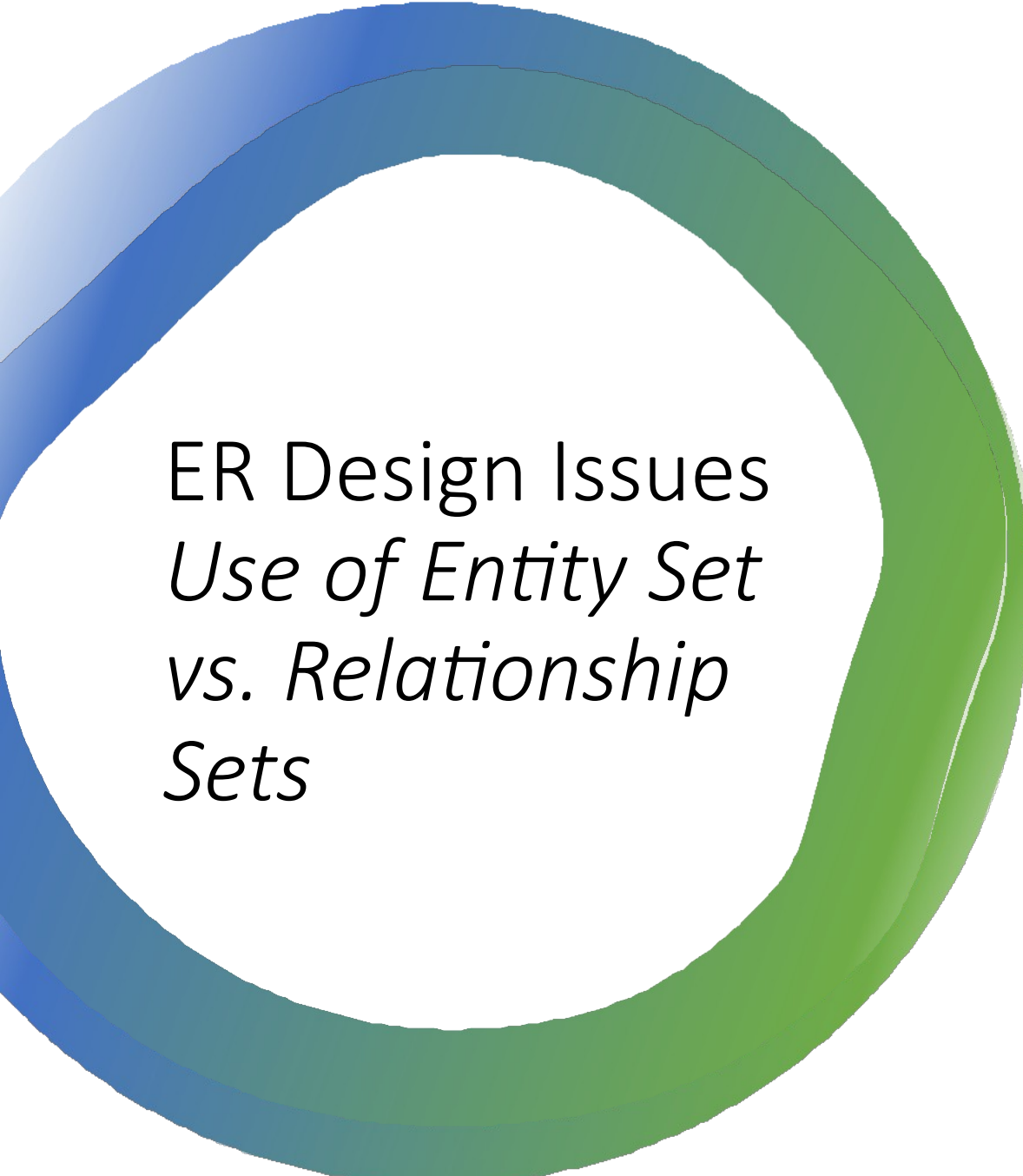
Use of Entity
Set vs.
Relationship
Sets

Placing
Relationship
Attributes



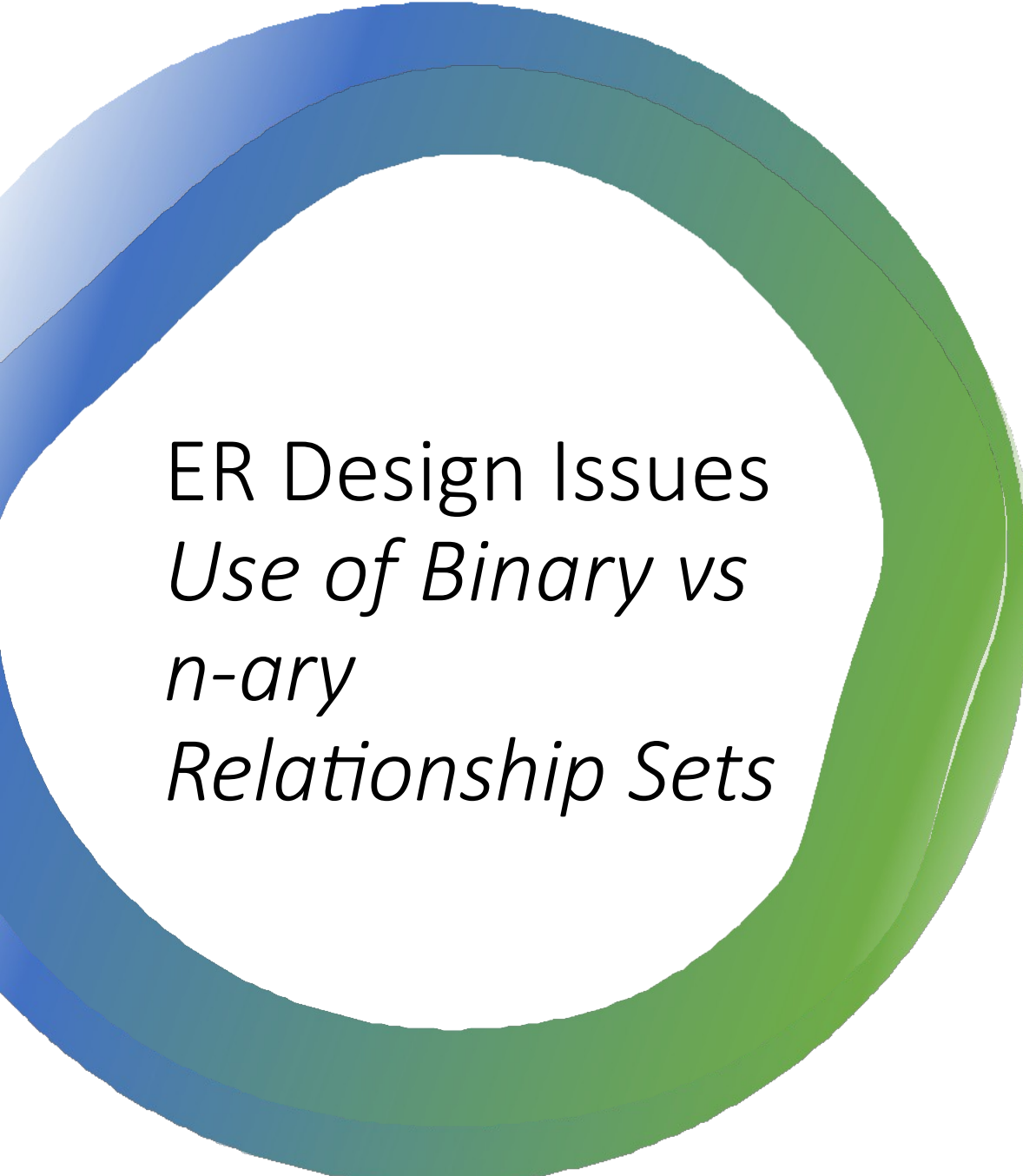
ER Design Issues *Use of Entity Set vs Attributes*

- The use of an entity set or attribute depends on the structure of the real-world enterprise that is being modelled and the semantics associated with its attributes.
- It leads to a mistake when the user use the primary key of an entity set as an attribute of another entity set.
- Instead, he should use the relationship to do so.
- Also, the primary key attributes are implicit in the relationship set, but we designate it in the relationship sets.



ER Design Issues *Use of Entity Set vs. Relationship Sets*

- It is difficult to examine if an object can be best expressed by an entity set or relationship set.
- To understand and determine the right use, the user need to designate a relationship set for describing an action that occurs in-between the entities.
- If there is a requirement of representing the object as a relationship set, then its better not to mix it with the entity set.



ER Design Issues *Use of Binary vs n-ary Relationship Sets*

- Generally, the relationships described in the databases are binary relationships. However, non-binary relationships can be represented by several binary relationships.
- For example, we can create and represent a ternary relationship 'parent' that may relate to a child, his father, as well as his mother.
- Such relationship can also be represented by two binary relationships i.e, mother and father, that may relate to their child.
- Thus, it is possible to represent a non-binary relationship by a set of distinct binary relationships.



ER Design Issues

Placing Relationship Attributes

- The cardinality ratios can become an affective measure in the placement of the relationship attributes. So, it is better to associate the attributes of one-to-one or one-to-many relationship sets with any participating entity sets, instead of any relationship set.
- The decision of placing the specified attribute as a relationship or entity attribute should possess the characteristics of the real-world enterprise that is being modelled.



ER Design Issues

Placing Relationship Attributes

- For example, if there is an entity which can be determined by the combination of participating entity sets, instead of determining it as a separate entity. Such type of attribute must be associated with the many-to-many relationship sets.
- Thus, it requires the overall knowledge of each part that is involved in designing and modelling an ER diagram. The basic requirement is to analyze the real-world enterprise and the connectivity of one entity or attribute with other.